Project: Worked examples
Title: Thin cylindrical shell under internal pressure
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the minimum thickness of a thin cylindrical shell under internal pressure, for circumferential and longitudinal stress. It adds corrosion allowance and a design factor, and compares the thickness provided with the requirement.

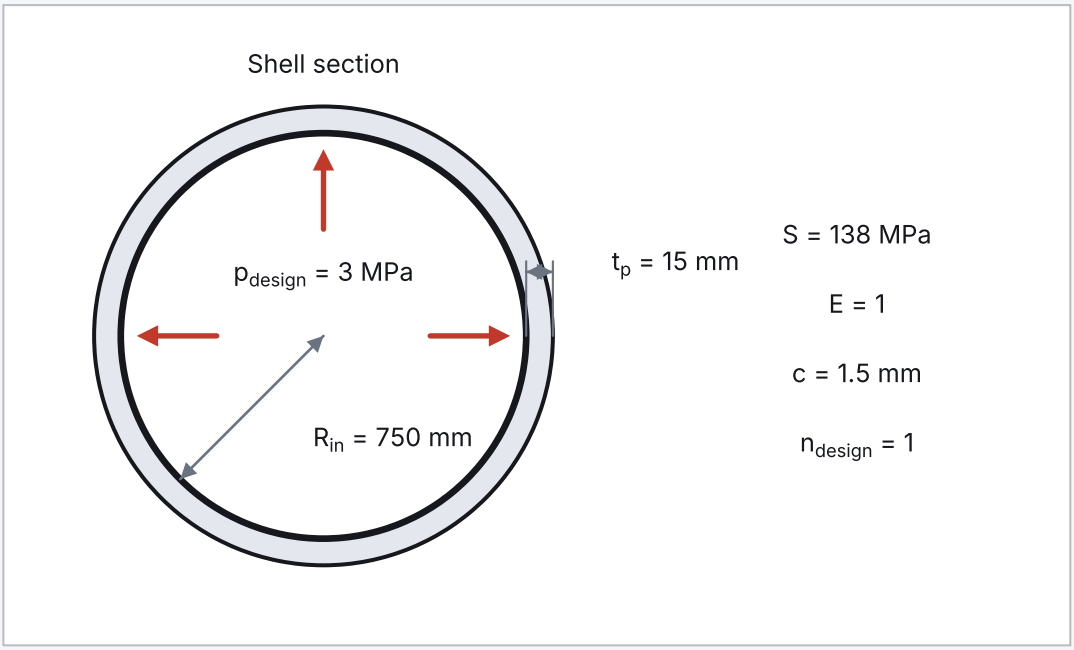


Figure 1 — A thin-walled cylinder in section under internal pressure: inside radius R_{in}, shell thickness t_p including the corrosion allowance c. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Thickness needed for the circumferential stress	$t_c = p_{design} \cdot R_{in} / (S \cdot E_j - 0.6 \cdot p_{design})$
Thickness needed for the longitudinal stress	$t_l = p_{design} \cdot R_{in} / (2 \cdot S \cdot E_j + 0.4 \cdot p_{design})$
Required thickness, with the design factor and corrosion allowance	$t_{req} = n_{design} \cdot \max(t_c, t_l) + c$
Hoop stress in the plate provided, after corrosion	$\sigma_\theta = p_{design} \cdot (R_{in} + 0.6 \cdot t) / (E_j \cdot t)$, at $t = t_p - c$
Range of validity of the thin-wall formulae	$t_c \le R_{in}/2$ and $p_{design} \le 0.385 \cdot S \cdot E_j$

Basis: Thickness of shells under internal pressure, circumferential stress[1](cl. UG-27(c)(1)); Thickness of shells under internal pressure, longitudinal stress[1](cl. UG-27(c)(2)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
R_{in}	Inside radius (corroded)	750	mm	drawing: DRG-PV-602 rev A	5 … 10000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
t_p	Shell thickness provided (nominal, before corrosion)	15	mm	drawing: DRG-PV-602 rev A	0.5 … 500 mm
S	Allowable stress at design temperature	138	MPa	spec: SPEC-MECH-03 design criteria	10 … 1000 MPa
p_{design}	Internal design pressure	3	MPa	spec: LOAD-SCHEDULE-01	0.01 … 100 MPa
c	Corrosion allowance	1.5	mm	spec: SPEC-MECH-03 design criteria	0 … 25 mm
E_j	Joint efficiency	1		spec: SPEC-MECH-03 design criteria	0.45 … 1
n_{design}	Design factor on the computed thickness (legacy; 1 for none)	1		spec: SPEC-MECH-03 design criteria	1 … 5

3. Assumptions

A1. The shell is cylindrical, away from heads, nozzles and supports, and carries membrane stresses only. (The scope of the UG-27 formulae.)
A2. The joint efficiency reduces the allowable stress uniformly. (Code-form simplification carried from the legacy module.)

4. Working

[8] $t_c = p_{design} * R_{in} / (S * E_j - 0.6 * p_{design})$ [1](VIII-1 UG-27(c)(1))
$= 3\ \text{MPa} \times 750\ \text{mm} / (138\ \text{MPa} \times 1 - 0.6 \times 3\ \text{MPa})$
$= 16.52\ \text{mm}$
Thickness for circumferential stress

[9] $t_l = p_{design} * R_{in} / (2 * S * E_j + 0.4 * p_{design})$ [1](VIII-1 UG-27(c)(2))
$= 3\ \text{MPa} \times 750\ \text{mm} / (2 \times 138\ \text{MPa} \times 1 + 0.4 \times 3\ \text{MPa})$
$= 8.1169\ \text{mm}$
Thickness for longitudinal stress

[10] $t_{req} = n_{design} * \max(t_c, t_l) + c$
$= 1 \times \max(16.52\ \text{mm}, 8.1169\ \text{mm}) + 1.5\ \text{mm}$
$= 18.02\ \text{mm}$
Required nominal thickness (design factor and corrosion allowance included)

[11] $t_{corr} = t_p - c$
$= 15\ \text{mm} - 1.5\ \text{mm}$
$= 13.5\ \text{mm}$
Corroded thickness provided

[12] $\sigma_h = p_{design} * (R_{in} + 0.6 * t_{corr}) / (E_j * t_{corr})$ [1](VIII-1 UG-27(c)(1))
$= 3\ \text{MPa} \times (750\ \text{mm} + 0.6 \times 13.5\ \text{mm}) / (1 \times 13.5\ \text{mm})$
$= 168.47\ \text{MPa}$
Hoop stress at the corroded thickness provided

[13] $u = t_{req} / t_p$
$= 18.02\ \text{mm} / 15\ \text{mm}$
$= 1.2013$
Required over provided thickness

5. Results

SYMBOL	VALUE	DESCRIPTION
t_{req}	18.02 mm	Required nominal thickness (governing)
σ_h	168.5 MPa	Hoop stress at the corroded thickness
t_c	16.52 mm	Thickness, circumferential stress
t_l	8.117 mm	Thickness, longitudinal stress
u	1.21	Thickness utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Provided thickness against required	18.02 mm	15 mm	1.21	[1](VIII-1 UG-27)

7. Applicability

None.

8. Verification

Template mech.thin-vessel@1.1.0 Content hash 4e7be0d562d46f01 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. American Society of Mechanical Engineers. Boiler and Pressure Vessel Code, Section VIII, Division 1: Rules for Construction of Pressure Vessels. ASME.